

Bingo Game - Complete Code Audit & Production Readiness Report

Project Overview

The project is a **Next.js 15 + React 19 + TypeScript + Supabase** Bingo application with:

- Host Dashboard
- Player Join Flow
- Lobby System
- Real-time updates using Supabase Realtime
- Number Bingo Mode
- Bollywood Bingo Mode
- Claim Validation
- Ticket Generation
- PWA support
- Zustand state management

Current Database Schema

games

Stores game configuration and state.

```
games
├── id
├── host_secret
├── game_type
├── number_range
├── dataset_size
├── status
├── game_name
├── host_name
├── max_players
├── auto_call
├── call_interval
├── ticket_size
├── game_code
├── auto_call_active
├── paused
└── created_at
```

Purpose

- Lobby management
 - Host authentication
 - Auto-call settings
 - Game state
-

players

```
players
┌ id
├ game_id
├ join_token
├ display_name
└ joined_at
```

Purpose

- Player registration
 - Join links
 - Player identity
-

player_tickets

```
player_tickets
┌ id
├ player_id
├ game_id
├ ticket_data (jsonb)
└ created_at
```

Purpose

Stores the generated bingo ticket.

called_items

```
called_items
┌ id
├ game_id
├ item_id (TEXT)
```

```
call_order
called_at
```

Purpose

Stores every called number, song, or movie.

player_marks

```
player_marks
id
player_id
item_id (TEXT)
marked_at
```

Purpose

Stores player markings.

claims

```
claims
id
game_id
player_id
claim_type
claim_data
is_valid
validation_reason
status
created_at
```

Purpose

Stores winner claims.

bollywood_mappings

```
bollywood_mappings
id
game_id
number
```

```
movie_name
image_url
dialogue
created_at
```

Purpose

Number → Bollywood mapping.

game_templates

```
game_templates
id
template_name
game_type
ticket_size
number_range
dataset_size
mappings_data
settings
created_at
```

Purpose

Reusable game presets.

Security Issues

1. RLS Policies Too Open

Found policies similar to:

```
USING (true)
WITH CHECK (true)
```

Risk

Anyone can:

- Create games
- Update games
- Insert claims
- Modify claims
- Insert called numbers

- Modify mappings

Priority

 Critical

2. Host Secret Exposed in URL

Current:

```
/host/{gameId}/{secretKey}
```

Example:

```
/host/123/SECRET_KEY
```

Anyone with the URL becomes the host.

Recommendation

Use:

- Supabase Auth
 - Magic Links
 - Host Sessions Table
-

3. No Rate Limiting

Potential abuse:

- Claim spam
- Join spam
- Mark spam

Recommendation

Move critical actions through:

- Next.js API Routes
 - Supabase Edge Functions
-

4. Client Controls Entire Game

Current browser can:

- Start game
- Pause game
- Resume game
- Call numbers
- Approve claims

Recommendation

Move authority to backend APIs.

Bugs & Logic Risks

1. Winner Race Condition

Scenario:

- Player A claims Full House
- Player B claims Full House

at nearly the same time.

Result

Multiple winners possible.

Solution

Create:

```
prize_locks

id
game_id
claim_type
winner_player_id
locked_at
```

2. Auto Call Stops on Refresh

Current implementation relies on browser timers.

Problem

Auto-call stops when:

- Host refreshes
- Host closes tab
- Internet disconnects

Solution

Store:

```
next_call_at
```

in database and recover state automatically.

3. Missing Host Reconnection Recovery

Need recovery for:

- Current number
 - Called history
 - Timer state
 - Pending claims
 - Auto-call status
-

4. Duplicate Player Names

Current system allows:

```
Deep  
Deep  
Deep  
Deep
```

Recommendation

Enforce uniqueness per game.

5. Unlimited Join Attempts

Need server-side validation of:

max_players

Missing Features Compared to Commercial Bingo Platforms

QR Code Join

Host screen should display:

- QR Code
 - Join Link
 - Game Code
 - Share Button
-

Live Leaderboard

Display:

- Rank
 - Player Name
 - Marks
 - Patterns Completed
-

Winner Celebration

When winner approved:

- Confetti
 - Fireworks
 - Winner Banner
 - Share Button
-

Multiple Prize Pools

Examples:

- Early Five
- Top Row
- Middle Row
- Bottom Row
- Corners

- X Pattern
 - Full House
 - Second Full House
 - Lucky Seven
-

Game History

Host can review:

- Previous games
 - Winners
 - Claims
 - Players
 - Called numbers
-

Spectator Mode

Allow non-playing viewers.

Host Notes

Host can attach:

- Rules
 - Instructions
 - Prize Information
-

Scheduled Games

Support:

- Future games
 - Scheduled start times
 - Invitations
-

Analytics Dashboard

Track:

- Total games
- Total players
- Average game duration
- Popular game modes

WhatsApp Sharing

Allow host to:

- Share Join Link
- Share QR Code
- Share Game Code

Recommended Database Improvements

game_events

```
id
game_id
event_type
payload
created_at
```

Tracks:

- joined
- left
- called
- claimed
- won
- paused
- resumed

prize_locks

```
id
game_id
claim_type
winner_player_id
locked_at
```

host_sessions

```
id
game_id
```

```
token
expires_at
```

game_results

```
id
game_id
winner_id
claim_type
created_at
```

Performance Assessment

Current Capacity

Suitable for:

```
10-50 players
```

Recommended Improvements for 100–500 Players

- Better indexing
 - Pagination
 - Event tables
 - Server-side validation
 - Optimized realtime subscriptions
-

For 1000+ Players

Consider:

- Dedicated backend service
 - Event-driven architecture
 - Redis caching
 - WebSocket gateway
-

Launch Readiness Checklist

Must Fix Before Launch

- Fix RLS policies
 - Remove host secret exposure
 - Add winner locking
 - Add server-side claim validation
 - Add auto-call persistence
 - Add reconnection recovery
 - Enforce max players
 - Prevent duplicate joins
 - Add rate limiting
-

Should Fix Before Launch

- QR Join
 - WhatsApp Sharing
 - Winner Celebration
 - Analytics Dashboard
 - Game History
-

Nice to Have

- Spectator Mode
 - Scheduled Games
 - Tournament Mode
 - Global Leaderboards
 - Multi-Room Management
-

Final Assessment

Area	Score
UI Structure	8.5 / 10
Game Logic	8 / 10
Database Design	8 / 10
Realtime Architecture	8 / 10
Security	4 / 10
Scalability	7.5 / 10
Production Readiness	7 / 10

Conclusion

The Bingo Game project is a strong beta-stage application with a solid architecture and good realtime design.

The primary focus before production launch should be:

1. Security hardening (RLS, host authentication, server-side validation)
2. Winner locking and race condition handling
3. Auto-call persistence and recovery
4. Reconnection handling
5. Rate limiting and abuse prevention

After these improvements, the project can evolve into a production-ready Bingo platform capable of handling large-scale public games.